UNIT 18 75 min

Visualizing Data with matplotlib

Choose the right chart, build line, bar, scatter, and histogram plots, label them honestly, and save them.

By the end of this unit you will be able to

- Choose a chart type that matches the question: comparison, trend, relationship, or distribution
- Draw a plot with `matplotlib.pyplot` and add a title, axis labels, and a legend
- Build several plots in one figure with `subplots`
- Plot directly from a pandas DataFrame
- Save a figure to a file and explain why axis choices can mislead

Pick the chart for the question

A chart answers one question. Start with the question, then choose the chart. The most common mistake is choosing a chart because it looks impressive rather than because it answers the question.

Question	Chart	Function
How do groups compare?	Bar chart	`plt.bar`
How does something change over time?	Line chart	`plt.plot`
Are two numbers related?	Scatter plot	`plt.scatter`
How are values spread out?	Histogram	`plt.hist`

Note

The plotting code in this unit is not run in the browser sandbox, because a sandbox cannot display a figure. Run it in Google Colab (free), in a Jupyter notebook, or as a script on your computer. Each example has been run with matplotlib installed.

Your first plot

first_plot.py

```
import matplotlib.pyplot as plt

months = ["Aug", "Sep", "Oct", "Nov"]
hours = [4, 6, 5, 7]

plt.plot(months, hours, marker="o")
plt.title("Study hours per month")
plt.xlabel("Month")
plt.ylabel("Hours")
plt.show()
```

Every chart needs a title, labeled axes with units, and a legend if it has more than one series. `plt.show()` opens the window. In a notebook, the figure appears below the cell.

Bar charts for comparison

bars.py

```
import matplotlib.pyplot as plt

periods = ["1", "2", "3", "4"]
averages = [86.5, 82.0, 90.2, 78.4]

plt.bar(periods, averages, color="#3b5bdb")
plt.title("Average score by period")
plt.xlabel("Period")
plt.ylabel("Average score (out of 100)")
plt.ylim(0, 100)
plt.show()
```

Watch out

Bar charts should start at zero. A bar that starts at 80 makes a 2-point difference look like a 50% difference. That is a misleading chart, even if every number is correct. Line charts and scatter plots are allowed to use a narrower range.

Scatter plots for relationships

scatter.py

```
import matplotlib.pyplot as plt

hours_studied = [1, 2, 3, 4, 5, 6, 7, 8]
test_scores = [55, 60, 66, 70, 74, 79, 85, 88]

plt.scatter(hours_studied, test_scores)
plt.title("Study time and test score")
plt.xlabel("Hours studied")
plt.ylabel("Test score")
plt.show()
```

A scatter plot shows whether two numbers move together. It does not prove that one causes the other. Say "is associated with" unless you have an experiment.

Histograms for distributions

hist.py

```
import matplotlib.pyplot as plt

scores = [55, 62, 68, 70, 71, 73, 75, 78, 79, 80, 82, 85, 88, 91, 95]

plt.hist(scores, bins=5, edgecolor="white")
plt.title("Distribution of test scores")
plt.xlabel("Score")
plt.ylabel("Number of students")
plt.show()
```

The number of bins changes the picture. Too few hides detail, and too many makes noise. Try a few values before you decide what the data shows.

Several plots in one figure

`plt.subplots(rows, cols)` returns a figure and a grid of axes. Draw on each axis with its own method: `ax.plot`, `ax.bar`, and so on. Use `fig.tight_layout()` to stop labels from overlapping.

subplots.py

```
import matplotlib.pyplot as plt

hours = [1, 2, 3, 4, 5]
scores = [55, 62, 66, 73, 79]

fig, (left, right) = plt.subplots(1, 2, figsize=(9, 3.5))
left.plot(hours, scores, marker="o")
left.set_title("Trend")
right.scatter(hours, scores)
right.set_title("Relationship")
fig.tight_layout()
plt.show()
```

Plotting a DataFrame

pandas DataFrames and Series have a `.plot` method that calls matplotlib for you. It is a fast way to look at data, but name your axes explicitly before sharing a chart.

df_plot.py

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.DataFrame({"period": [1, 2, 3], "score": [86, 82, 90]})
ax = df.plot(x="period", y="score", kind="bar", legend=False)
ax.set_ylabel("Average score")
plt.tight_layout()
plt.show()
```

Saving a figure

`plt.savefig("chart.png", dpi=150)` writes the figure to a file. Call it *before* `plt.show()`, because `show` can clear the figure. Use PNG for slides and SVG or PDF for documents.

Judgment checks

- **Bars start at 80 instead of 0.** Exaggerates small differences, because the eye compares bar heights.
- **A scatter plot shows study hours and scores rising together.** That is an association. Other factors can drive both numbers.
- **A histogram with 2 bins for 40 scores.** Hides the shape, because different parts of the distribution are merged.

Practice

▼ **Exercise 1 — Labeled trend**

Plot the list `[3, 5, 4, 8, 7]` against the days `["Mon", "Tue", "Wed", "Thu", "Fri"]` as a line with markers. Add a title, both axis labels, and save it to `trend.png`.

Discussion

1. Find a chart in a newspaper or on social media. What question does it answer? Would a different chart answer it better?
2. When is a truncated axis acceptable? Give one example where it is and one where it is not.
3. What would you need to know before claiming that a scatter plot shows a cause?

Key terms

matplotlib

The standard Python library for drawing charts. Its `pyplot` module is usually imported as `plt`.

Figure

The whole picture that matplotlib draws, which can hold one or more axes.

Axes

A single plotting area inside a figure, with its own x and y axes, title, and data.

Histogram

A chart that groups numeric values into bins and shows how many fall in each bin.

Bin

A range of values grouped together in a histogram.

Scatter plot

A chart of points, one per observation, that shows the relationship between two numbers.

Truncated axis

An axis that does not start at zero, which can exaggerate differences between bars.

subplots

A function that creates a grid of axes in one figure.

See all terms in the glossary →